

## Question 3 a

```
install.packages("GGally")
```



Installing package into '/usr/local/lib/R/site-library'  
(as 'lib' is unspecified)

also installing the dependencies 'patchwork', 'ggstats', 'plyr'

```
install.packages("faraway")
```



Installing package into '/usr/local/lib/R/site-library'  
(as 'lib' is unspecified)

also installing the dependencies 'minqa', 'nloptr', 'RcppEigen', 'lme4'

```
# Load necessary libraries
library(pander)
library(ggplot2)
library(GGally)
library(faraway)

# Load the dataset
data(teengamb, package="faraway")

# Plot pairwise relationships
ggpairs(teengamb)

# Fit the linear model
lm.fit <- lm(gamble ~ sex + status + income + verbal, data=teengamb)

# Display model summary
summary(lm.fit)

# Print regression coefficients
coefficients <- coef(lm.fit)
print(coefficients)
```



Attaching package: 'faraway'

The following object is masked from 'package:GGally':

happy

Call:

```
lm(formula = gamble ~ sex + status + income + verbal, data = teengamb)
```

Residuals:

| Min     | 1Q      | Median | 3Q    | Max    |
|---------|---------|--------|-------|--------|
| -51.082 | -11.320 | -1.451 | 9.452 | 94.252 |

Coefficients:

|             | Estimate  | Std. Error | t value | Pr(> t )     |
|-------------|-----------|------------|---------|--------------|
| (Intercept) | 22.55565  | 17.19680   | 1.312   | 0.1968       |
| sex         | -22.11833 | 8.21111    | -2.694  | 0.0101 *     |
| status      | 0.05223   | 0.28111    | 0.186   | 0.8535       |
| income      | 4.96198   | 1.02539    | 4.839   | 1.79e-05 *** |
| verbal      | -2.95949  | 2.17215    | -1.362  | 0.1803       |

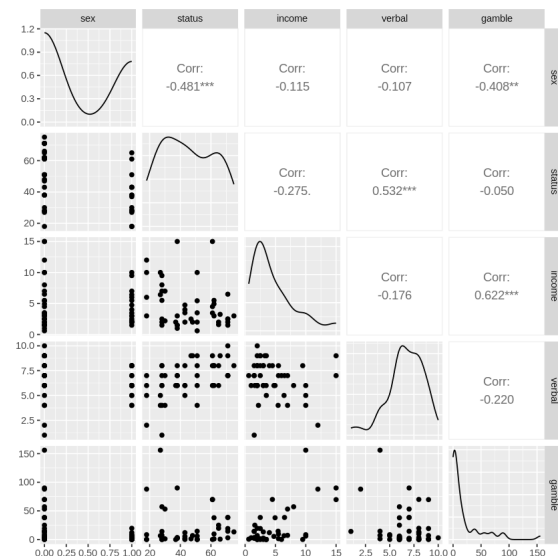
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.69 on 42 degrees of freedom

Multiple R-squared: 0.5267, Adjusted R-squared: 0.4816

F-statistic: 11.69 on 4 and 42 DF, p-value: 1.815e-06

|             | sex         | status       | income     | verbal      |
|-------------|-------------|--------------|------------|-------------|
| (Intercept) | 22.55565063 | -22.11833009 | 0.05223384 | 4.96197922  |
|             |             |              |            | -2.95949350 |



### Question 3 b

```
# Fitted values and residuals from the model
```

```
fitted_values <- fitted(lm.fit)
```

```
residuals <- residuals(lm.fit)
```

```
# Sum of squares of residuals (SSR)
```

```
ssr <- sum(residuals^2)
```

```
# Sample size (n) and number of predictors (p)
```

```
n <- length(residuals)
```

```
p <- length(coef(lm.fit))
```

```
# Unbiased estimate of variance
```

```
unbiased_est <- ssr / (n - p)
```

```
print(unbiased_est)
```



```
[1] 514.8516
```

### Question 3 c

```
# Proportion of variance explained by the predictors
```

```
var.explained.proportion <- summary(lm.fit)$r.squared
```

```
var.unexplained.proportion <- 1 - summary(lm.fit)$r.squared
```

```
# Display proportion of variance explained
pander(data.frame(var.explained.proportion= var.explained.proportion), caption="Proportion of Variance Explained")
```



| var.explained.proportion |
|--------------------------|
| 0.5267                   |

Table: Proportion of Variance Explained

Question 3 d // the sample correlation between the residuals and the fitted values in regression models is always zero, this is due to minimizing sum of squared residuals.

Question 3 e Zero correlation, if Income is a predictor it is already accounted by model.

Question 3f

### Simple Explanation:

beta 1 (coefficient for the variable "sex") will tell us how much more or less males spend on gambling compared to females. if beta 1 is greater than zero, males spend more than female and vice versa

Question 4

Question 4c // The first equation represents the full model, in second equation a condition is put excluding a predictor. This can remove a variable that might be important in the model, that's why model's performance will be reduced and sum of squared residuals will be larger. So first equation will be less than second.

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```
# Load necessary data and libraries
data(prostate, package="faraway")
library(broom)

# Create a dataframe to hold the results of the experiment
model.stats <- data.frame(num.predictors=integer(), r.squared=numeric(), residual.se=numeric(), model.string=character())

# Fit the first model with 1 predictor
lm.fit <- lm(lpsa ~ lcavol, data=prostate)
summary(lm.fit)
model.summary <- summary(lm.fit)

# Extract statistics
r.squared <- model.summary$r.squared
residual.se <- model.summary$sigma
model.string <- "lpsa ~ lcavol"

# Append results to the dataframe
model.stats <- rbind(model.stats, list(num.predictors = 1, r.squared=r.squared, residual.se=residual.se, model.string=model.string))

# Fit models with increasing number of predictors
predictor.combinations <- list(
  "lpsa ~ lcavol + lweight",
  "lpsa ~ lcavol + lweight + svi",
  "lpsa ~ lcavol + lweight + svi + lbph",
  "lpsa ~ lcavol + lweight + svi + lbph + age",
  "lpsa ~ lcavol + lweight + svi + lbph + age + lcp",
  "lpsa ~ lcavol + lweight + svi + lbph + age + lcp + pgg45",
  "lpsa ~ lcavol + lweight + svi + lbph + age + lcp + pgg45 + gleason"
)

# Loop through each model combination
for (i in seq_along(predictor.combinations)) {
  lm.fit <- lm(as.formula(predictor.combinations[[i]]), data=prostate)
  model.summary <- summary(lm.fit)

  # Extract statistics
  r.squared <- model.summary$r.squared
  residual.se <- model.summary$sigma
  model.string <- predictor.combinations[[i]]

  # Append results to the dataframe
```

```
model.stats <- rbind(model.stats, list(num.predictors = 1 + 1, r.squared = r.squared, residual.se = residual.se, model.string = model.string))

# Convert the model strings to character
model.stats$model.string <- as.character(model.stats$model.string)

# Load pander for table display
install.packages("pander")
library(pander)

# Display the model stats
rownames(model.stats) <- NULL
pander(model.stats, caption = "Model Stats")

# Plot the relationship between Residual SE and R-squared
install.packages("ggplot2")
install.packages("GGally")
library(ggplot2)
library(GGally)

p <- ggplot(model.stats, aes(x = residual.se, y = r.squared)) + geom_point()
p <- p + geom_text(aes(label = num.predictors, hjust = 1.5, vjust = 1.5))
p + ggtitle("Residual SE vs R-squared with Number of Predictors") + theme(plot.title = element_text(hjust = 0.5))
```



```
Call:
lm(formula = lpsa ~ lcavol, data = prostate)

Residuals:
    Min       1Q   Median       3Q      Max
-1.67625 -0.41648  0.09859  0.50709  1.89673

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.50730     0.12194   12.36  <2e-16 ***
lcavol       0.71932     0.06819   10.55  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7875 on 95 degrees of freedom
Multiple R-squared:  0.5394,    Adjusted R-squared:  0.5346
F-statistic: 111.3 on 1 and 95 DF,  p-value: < 2.2e-16
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```